

Subject: Data science

- ① C
- ② C
- ③ D
- ④ B
- ⑤ A
- ⑥ C
- ⑦ B
- ⑧ D
- ⑨ A
- ⑩ A
- ⑪ B
- ⑫ C
- ⑬ B
- ⑭ C
- ⑮ B
- ⑯ C

(17)

B

(18)

C

(19)

B

(20)

B

(21)

supervised learning is a type machine learning algorithm that use labelled data to train the machine learning dataset.

(23)

Regression analysis is used for supervised learning to find continuous value of dependent variable by using one or more independent variable.

(24)

- handling missing value
- outlier detection

(25)

- colab
- Jupyter

(26)

structured data

- high good organization
- Ex - Excel, csv,
- easily understand by computers

semi-structured data

- not really good organization
- Ex. audio, html, XML
- need to pre-do data pre-processing before make computer understand

(27) lower = $Q_1 - 1.5(IQR)$

higher = $Q_3 + 1.5(IQR)$

(28) Data pre-processing is the process of handling the data before pass it on to ^{computer} machine learning algorithm to do the prediction or analysis such as data profiling, data cleansing...

(29) because Data normalization is used to scale ~~the~~ huge data so, it can be presented in a smaller range.

(31) K is the constant that ~~is~~ it can be represented as the ^{number of} group or cluster of those ~~unlabeled~~ data points.

(34) - The 1st classifier is the best the other classifier.

because correct data classification

and spaces the ~~far~~ lines far from each other.

- The ~~send~~ 2nd classifier is good for distance the lines but fail for data classification

- The 3rd classifier is good for data classification but but the space between two lines are too closed.

(30) continuous value can be any value, Example 1.47

we can discretize them by look at value behind the (.) if it is below 5, it will be 1. if it is above 5 it will be 2.

(29) Model M_1 is not good since the accuracy between training and testing dataset didn't come with acceptable Value.

- (32)
- Linear Regression
 - Logistic Regression
 - support Vector machine
 - Random Forest
 - Decision Tree.

- (35)
- try to classify data classification correctly
 - try to keep space between the two lines as far as possible.